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Detecting binary supermassive black holes in quasars with UV spectropolarimetry

Julie Biedermann [1]

Introduction

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  Relativistic outflow of ionized matter emerging in the polar axis sometimes extending to megaparsec scales and forming large radio lobes.
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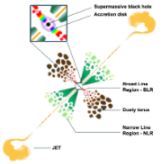
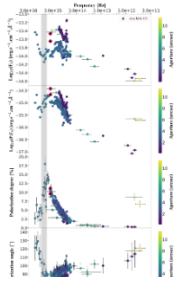


Figure 1 : Unscaled schema of AGN strcuture, *Marin (2026)*

Studying quasars offers a unique opportunity to probe the physical processes in the immediate surroundings of SMBHs, where complex geometries dominate. This geometry is schematically illustrated in Figure 1 (see right).

Markarian 231 (Mrk 231) - nearest quasar known



Mrk 231, at redshift z = 0.0422, is one of the brightest ultraluminous infrared galaxies (ULIRGs) in the local universe. The presence of disturbed morphology and tidal features indicate that this quasar is likely at an advanced stage of galactic merger, and is therefore a promising candidate for hosting an SMBBH system. A detailed analysis of its UV-optical spectrum revealed atypical spectral features.

Unsual UV-optical spectral shape - See Figure 2 top panel :

- Pronounced cutoff in the UV continuum around 3000 Å, followed by a slow rise below 2500 Å, deviates significantly from typical quasar spectrum - interpreted as a spectral signature of a binary accretion system, where the secondary BH opens a gap in the accretion disk (*Yan et al 2015*).

Method - SED of binary SMBH systems

Objective :

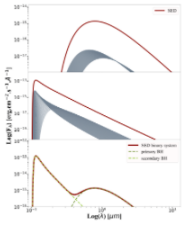
We aim to reproduce the total spectral energy distribution (SED) of Mrk 231 using the hypothesis of a binary supermassive black hole, which proposes a primary heavier SMBH with its accretion disk emitting in the optical and then we have the secondary lighter one inside the primary disk emitting mainly in the UV.

- **Modeling the Primary SMBH :**

We began by modeling the accretion disk of the first primary black hole using the model of Sakura and Sunyaev, which describes a geometrically thin and optically thick disk. THe emitted spectrum is treated as a multi-temperature blackbody.

- Inner radius :
  The inner radius of the accretion disk around the primary SMBH, $r_{int,p}$, is definied using the *Hill radius*, R_{Hill}, which depends on the the semimajor axis, a_{BBH}, and the mass ratio, q. This definies the limit within which the gravitatoinal influence of the primary SMBH dominates the secondary one.
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The SED for the primary SMBH emission is shown [illegible]red solid line in



Figuure 3 : ***Top***: SED of the primary (more massive) SMBH in red. Each of the 100 emitting radii are shown in gray. ***Middle***: Same plot, but for the secondary (lighter) SMBH. ***Bottom***: Combined SED of the two SMBHs in a binary system based on the model proposed by *Yan et al. (2015)*, with $M_{*,p} = 1.5 \times 10^8$ $M_\odot$ for the primary SMBH, and $M_{*,s} = 4.5 \times 10^6$ $M_\odot$ for the secondary SMBH. (*Biedermann et al. 2025*)

Method - Fitting the polarized flux data

We analyzed the polarimetric response of our SED model without modifying the geometrical configuration of the system.

Fitting the polarized flux data

Each component of the AGN (primary and secondary SMBHs, dusty torus, host galaxy) was assigned a wavelength-independent linear polarization, described by Stokes parameters (I, Q, U, V).

The total intensity I_i was fixed by the best-fit flux $F_{\lambda,i}$ from the previous SED model. The associated linear polarization components were computed as :

- $Q_i = q_i \times I_i$
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where (q_i, u_i) are constant polarization fractions constrained to [–1, 1] for each component of the quasar. We fitted the total Stokes Q and U to the observed polarization of Mrk 231 in order to retrieve the best (q_i, u_i) for each component.

From the summed Stokes parameters, we derived the best fit model display in the Fugure 5 in the Results section :

- Polarized flux (top-right)
- Degree of polarization (bottom-left)
- Angle of polarization (bottom-right)

Results

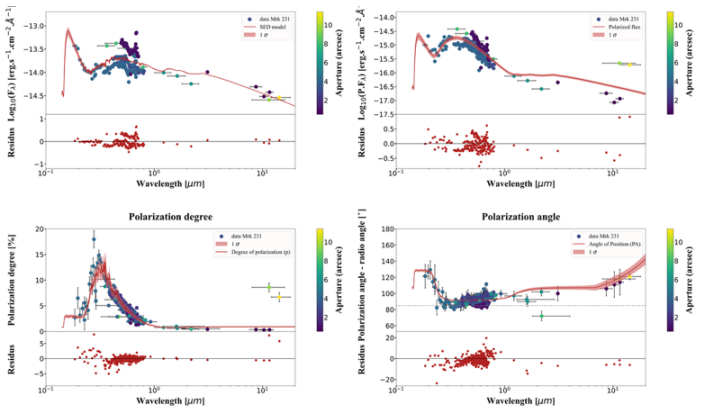


Figure 5 : ***Top left panel*** : Mrk~231 total flux as a function of wavelength. ***Top right panel*** : Polarized flux as a function of wavelength, established from the multiplication of the flux and the polarization degree. ***Bottom left panel*** : Degree of polarization as a function of wavelength. ***Bottom right panel*** : Polarization position angle minus the radio position angle as a function of wavelength. Below each panel, we give the residuals between the observational data and the theoretical model (*Biedermann et al. 2025 - accepted*).

Conclusion

Our results support the **binary SMBH hypothesis for Mrk 231**, as it successfully reproduces three independent observables: *the total flux, the degree of polarization, and the polarization angle*

The model also predicts a **second bump in the far-UV** in the polarized flux, originating from the lighter secondary black hole. However, the current lack of spectropolarimetric data in the far-UV prevents a direct confirmation of this prediction.

This study thus opens new perspectives for the exploration of quasar inner structures through **space-based UV polarimetry**, in particular with upcoming instruments such as **POLLUX/HWO**.

- Hight spectral resolution (R > 20 000) in the Lyα region to monitor potential variability
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Acknowledgements

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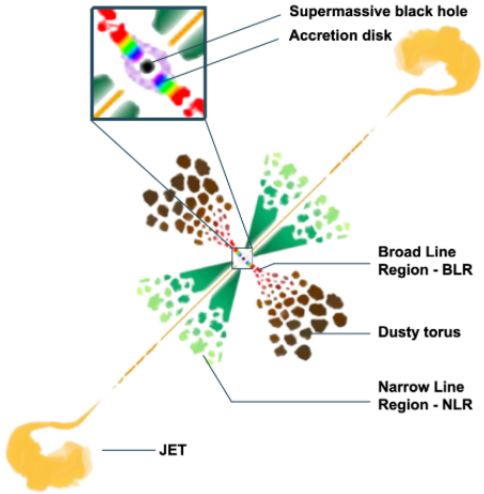


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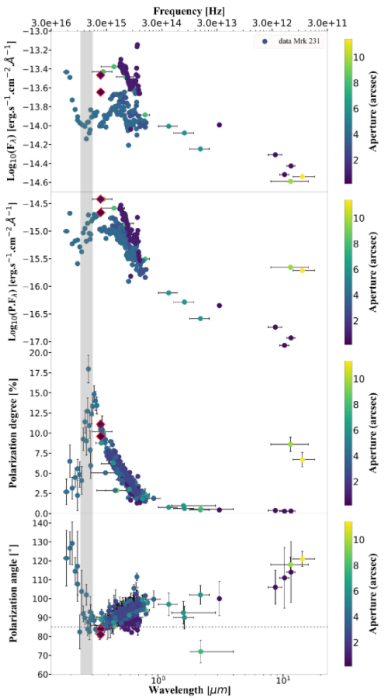


Figure 2 : ***Top panel*** : Reddening-corrected total flux of Mrk 231. ***Second panel*** : Polarized flux of Mrk 231, established from the multiplication of the flux and the polarization degree. ***Third panel*** : Linear polarization degree. ***Bottom panel*** : Polarization position angle of Mrk 231. The aperture is color-coded (in arcseconds). The gray zone represents the waveband in which we observe the cut-off in the UV spectrum, associated with a strong rise in the degree of polarization and a distinctive rotation in the polarization position angle. The data highlighted in red as the new ones we derived from the poorly explored HST/FOC observation of Mrk 231. (*Biedermann et al. 2025 - accepted*)

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Unique UV- optical polarization properties

- Degree of polarization reaches ~ 20% in the UV, then drops to less than 0.5% in the IR (See third panel on Figure 2).
- Smooth and continuous rotation in the polarization position angle from ~130° to ~80° and then ~100° from the UV to the IR suggesting a complex scattering geometry.

To study the intrinsic polarization properties of Mrk 231, we collected archival data with both total and polarized flux measurements across the UV to mid-IR range. As shown in Figure 2, the total flux spectrum exhibits the characteristic UV drop of Mrk 231, while the polarized flux shows a distinct peak at shorter wavelengths. The degree of polarization increases sharply around 3000 Å before decreasing toward the IR, and the polarization angle rotates smoothly with wavelength. These features confirm the intrinsic, wavelength-dependent nature of polarization in this AGN. In this study, we aim to test the binary SMBH scenario for Mrk 231 by modeling its total and polarized fluxes using an analytical approach, and comparing the predictions to UV spectropolarimetric observations.

Method - SED of binary SMBH systems

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The SED for the primary SMBH emission is shown by a red solid line in the top panel of the figure 3 (see right).

- **Moddeling the seconary SMBH :**

The secondary, lower-mass SMBH comes perturbs the primary disk, opening an inner gap, forming a secondary mini disk.

  - Inner radius :
    THe inner radius of the secondary SMBH corresponds to the ISCO (innermost stable circular orbit) for a non-rotating black hole.

  - Outer radius :
    THe outer radius od the secondary disk is definied as a fraction of the Roche lobe radius depending on the mass ratio q and the semi-major axis a_{BBH}.

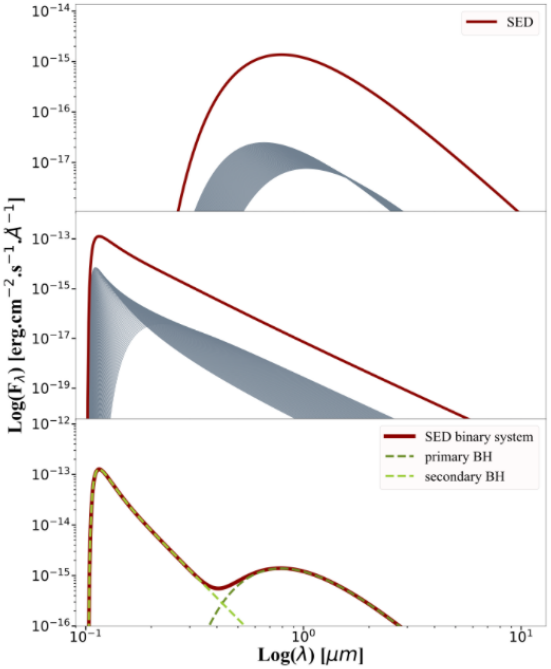
THe SED of the secondary SMBH is represented by a red solid line in the middle panel of the figure 3 (see right)

- **Combined SED :**

The total SED resulting from the combined emission from both accretion disks is shown by the red solid line in the bottom panel of the figure 3.

The first peak in the far-UV corresponds to the secondary SMBH with a mass of $M_{*,s} = 4.5 \times 10^6\ M_\odot$.

The second peak in the UV-optical corresponds to the primary SMBH emission with a mass of $M_{*,p} = 1.5 \times 10^8\ M_\odot$.



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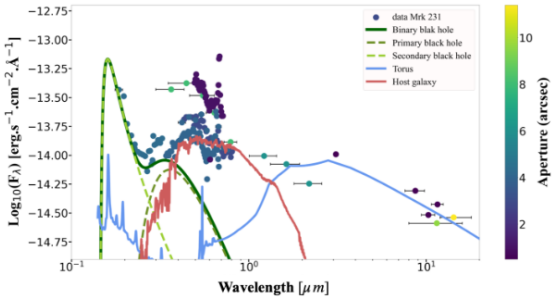


Figure 4 : Logarithm of the total flux as function of the wavelength of Mrk 231. The dark green line corresponds to the modeled SED of the binary SMBH system, with the light green dotted lines denoting the secondary SMBH ($M_{*,s} = 11 \times 10^6 M_\odot$) and the darker green dotted one is for the primary SMBH ($M_{*,p} = 1.6 \times 10^8\ M_\odot$). The light red line is for the host galaxy and the blue one is for the torus component.

- **Additonal components**

To construc our complete and comprhensive SED model and reproduce Mrk 231'emissoin, we incorporate additonal compnents associated with the strucure of the quasar.

  - **Host galaxy**
    We included the optical emissoin from the host galaxy using a 13 Gyr-old elliptical galaxy template from the work of *Polleta et al (2007)*.

  - **Dusry torus**
    We included the infrared contribution from the dusty torus surronding the binary SMBH system usinf the SED library of simulations provided by *Siebenmorgen et al (2015)*.

The final SED results from the combination of the binary SMBH model, the dusty torus and the host galaxy emission and is represented by the red line in the top left panel of the figure 5 (see Resultats section).

EQUATIONS

Outer secondary radius

$$R_{RL}(q) \approx \frac{0.49\ a_{BBH}\ q^{\frac{2}{3}}}{0.6\ q^{\frac{2}{3}} + log(1+q^{\frac{1}{3}})}$$

$$\rightarrow r_{out,s} = R_{RL}(q) \cdot f_{r,s}$$

Inner primary radius

$$R_{Hill} \sim a_{BBH}\left(\frac{M_{*,s}}{3M_{*,p}}\right)^{\frac{1}{3}}$$

$$\rightarrow r_{int,p} = R_{Hill} + \frac{a_{BBH}}{(1+q)}$$

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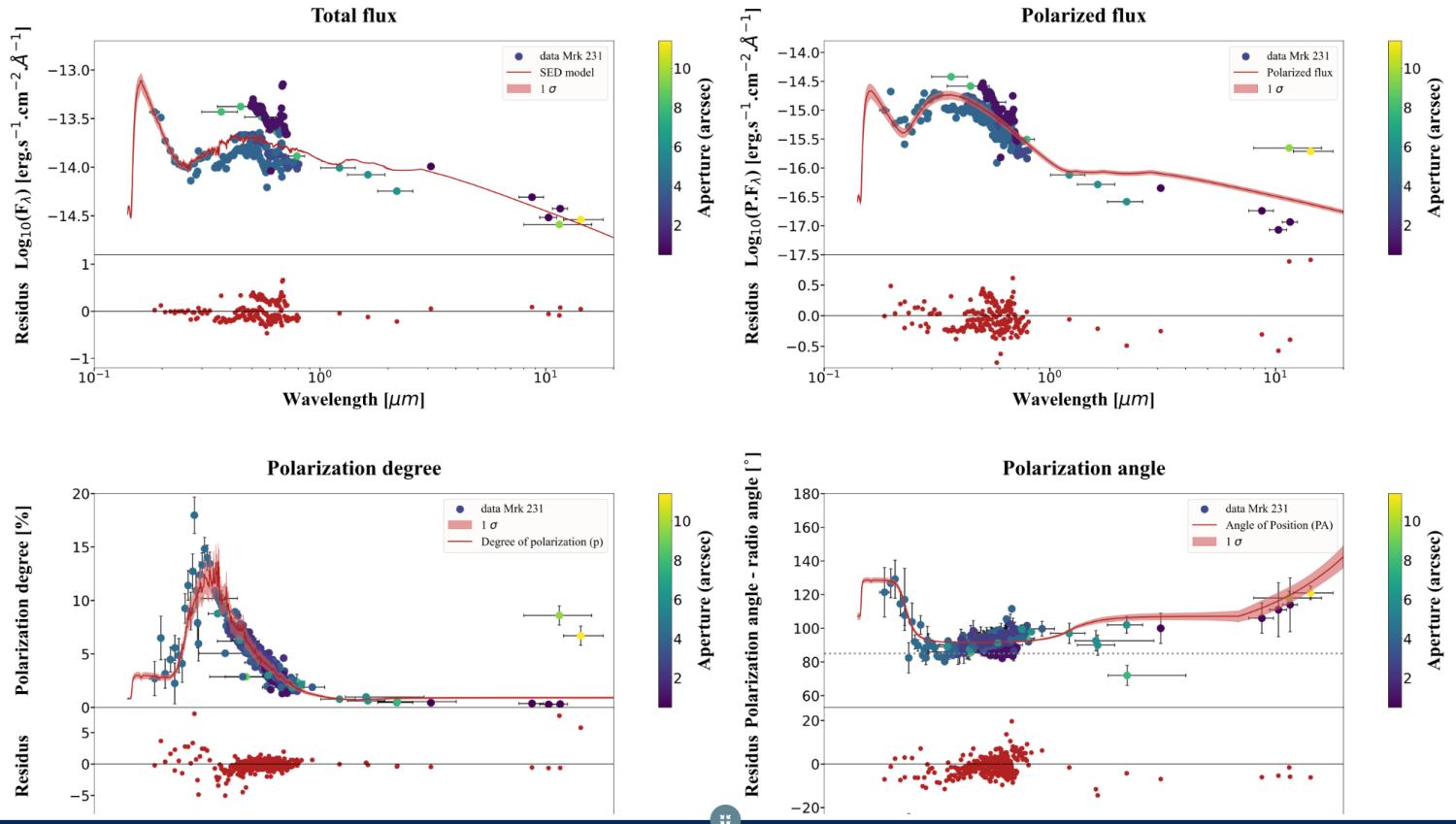
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Results

Figure 5 shows the **best-fit models** for both the **total flux** and the **polarization** of Mrk 231. The model predicts a **second bump** in the UV in the polarized flux, which corresponds to the emission from the lighter secondary supermassive black hole in our binary hypothesis.



Conclusion

Our results support the **binary SMBH hypothesis for Mrk 231**, as it successfully reproduces three independent observables: *the total flux, the degree of polarization, and the polarization angle*

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